CS151 Intro to Data Structures

Iterators
Recursion

Binary Search

warmup: what is printed?

```
List<String> words =
     new ArrayList<>(Arrays.asList("apple", "banana", "cherry", "date", "elderberry"));

Iterator<String> it = words.iterator();
while (it.hasNext()) {
    String word = it.next();
    if (word.length() % 2 == 0) {
         it.remove();
    }
}

System.out.println(words);
```

Announcements

- Lab5 and HW3 due Wednesday! (March 4th)
  - Stacks and queues

- chance to resubmit assignments over spring break...

- Lab6 and HW4 are released
  - due Monday March 16th

- Midterm Wednesday after break (March 18th)

Outline

- Iterators
- Recursion Review
- Binary Search

Iterators

- represents a sequence of elements and provides a way to iterate, or traverse, through those elements one at a time

`Iterable` versus `Iterator`?

- `Iterable`
  - `java.lang`
  - override `iterator()`
  - Doesn't store the iteration state
  - Removing elements during iteration isn't allowed
- `Iterator`
  - `java.util`
  - Override `hasNext()`, `next()`
  - Optional `remove()`
  - Stores iteration state (list cursor)
  - Removing elements during iteration supported

Let’s make ExpandableArray iterable

Outline

- Iterators
- **Recursion**
- Binary Search

Recursive functions – base case

Conditional statement that prevents infinite repetitions

Usually handles cases where:

- input is empty
- problem is at its smallest size

Recursion Example – Contains letter

Write a method called “containsLetter” that determines if a String contains a given character

Question: What are the parameters?

1. The character to look for
2. The string to be looking in

Question: What is the return type?

Code it!

Recursion Visualization – Contains letter

```
contains(“l”, “apple”) =
    contains(“l”, “apple”)
        contains(“l”, “pple”)
            contains(“l”, “ple”)
                contains(“l”, “le”)
                    return true
```

Recursive Method

Break problem down into smaller subproblem that we can repeat

Base case(s):

- no recursive calls are performed
- every chain of recursive calls must reach a base case eventually

Recursive calls:

- Calls to the same method in a way that progress is made towards a base case
- Often called “the rule”

Outline

- iterators
- Recursion
- **Binary Search**

Binary Search

- efficient search in a sorted list
- can be implemented **recursively**

Search steps:

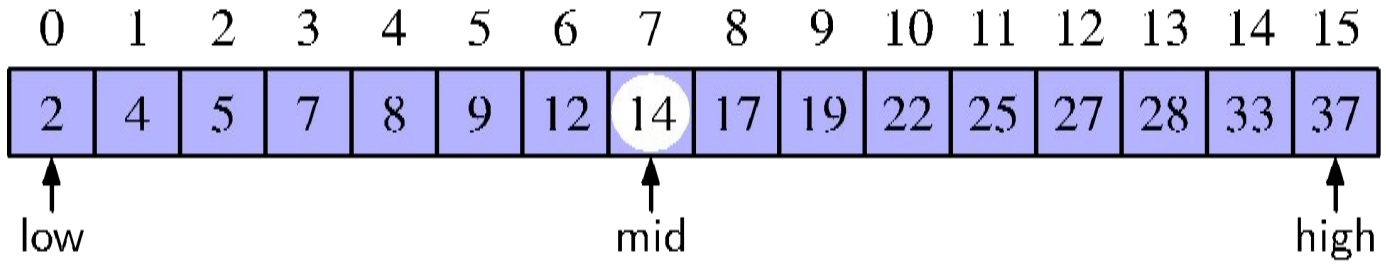
1. Calculate midpoint
2. Compare the value at the midpoint with the target value
   a. if equal:
      i. return index
   b. if target value < midpoint value:
      i. search the left portion of the list
   c. if target value > midpoint value:
      i. search the right portion of the list

Binary Search

Search for an integer (22) in an ordered list

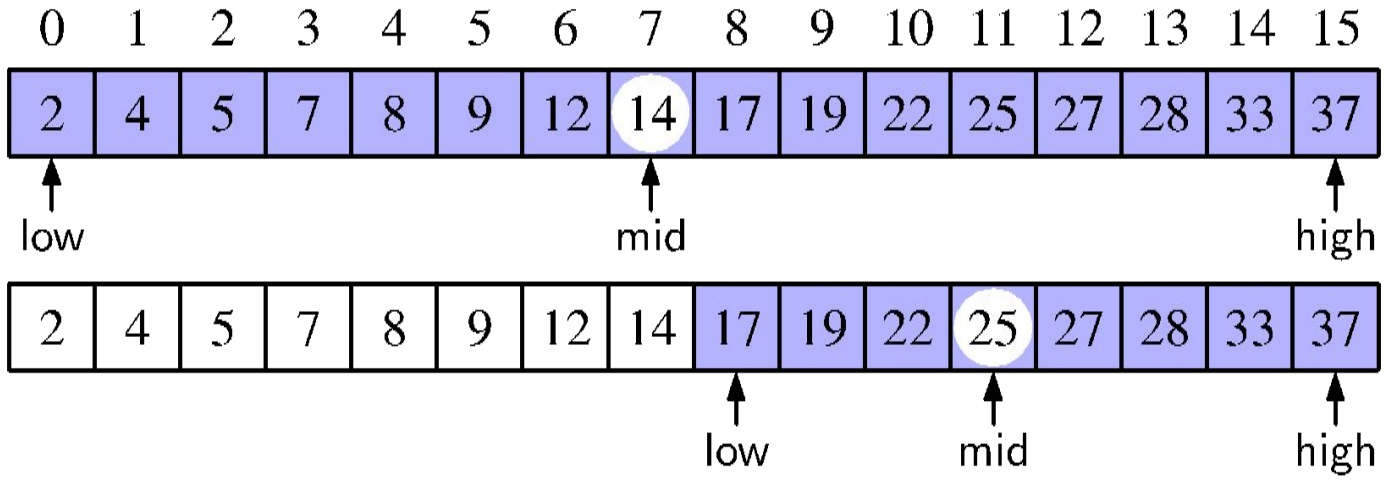
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
2	4	5	7	8	9	12	14	17	19	22	25	27	28	33	37

target = 22



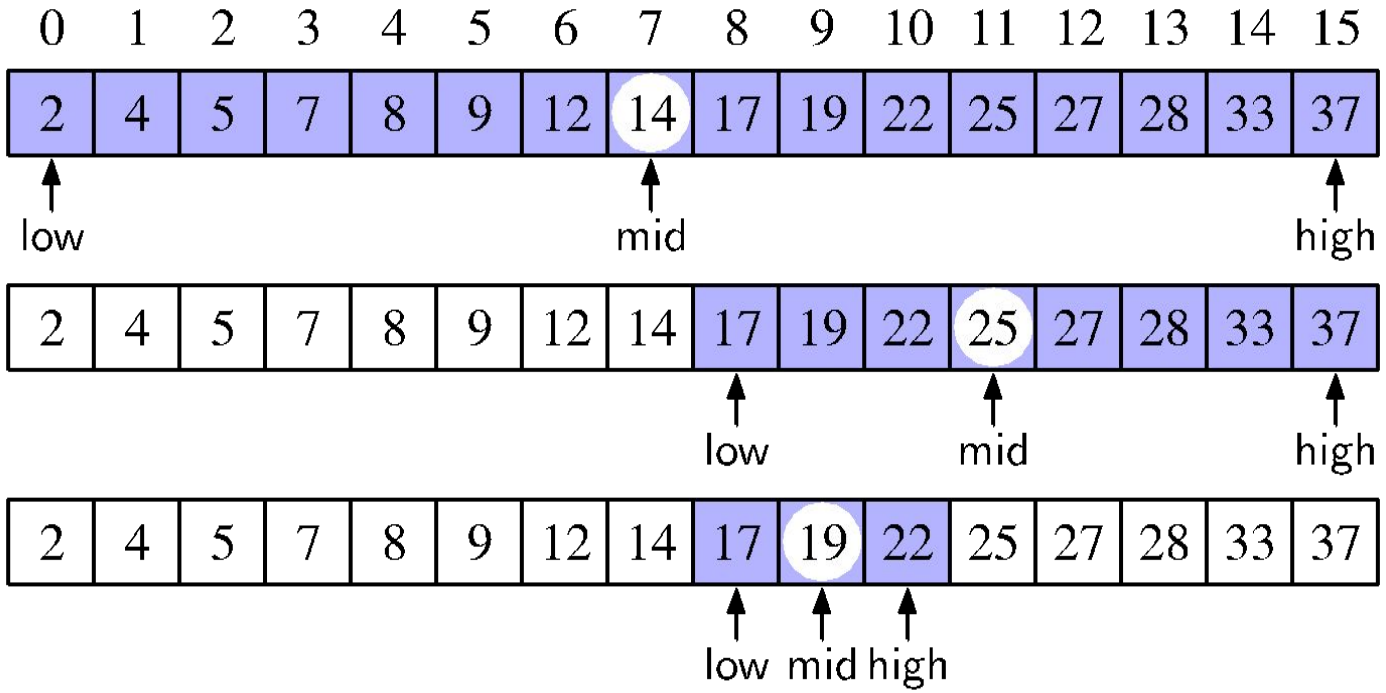

```
int mid = (high + low) / 2; //index
int midVal = arr[mid]; //14
```

target = 22

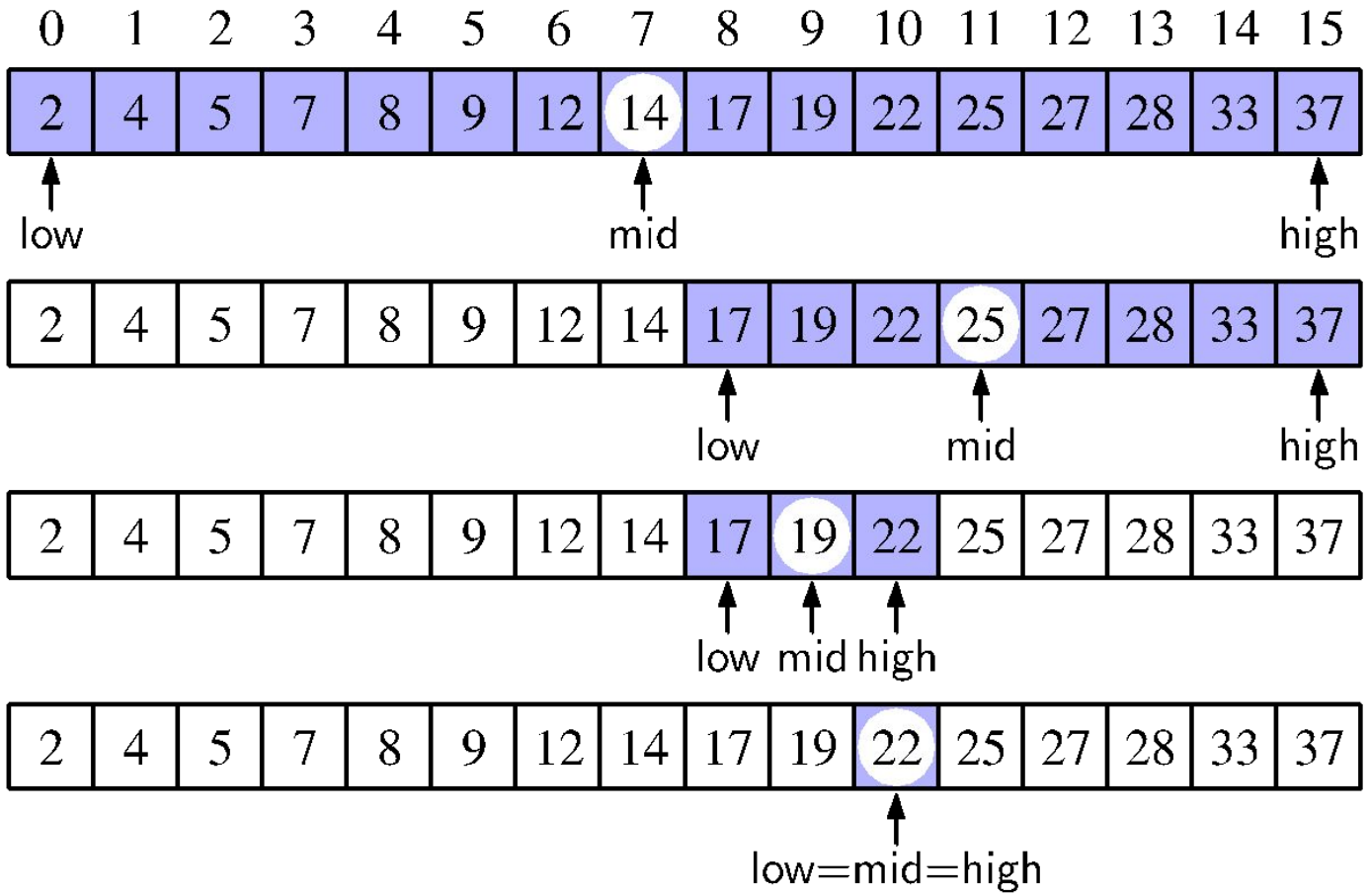


low = mid+1;

target = 22


0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
2 4 5 7 8 9 12 14 17 19 22 25 27 28 33 37
low
mid
high
2 4 5 7 8 9 12 14 17 19 22 25 27 28 33 37
low
mid
high
2 4 5 7 8 9 12 14 17 19 22 25 27 28 33 37
low
mid
high

target = 22


0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
2 4 5 7 8 9 12 14 17 19 22 25 27 28 33 37
low
mid
high
2 4 5 7 8 9 12 14 17 19 22 25 27 28 33 37
low
mid
high
2 4 5 7 8 9 12 14 17 19 22 25 27 28 33 37
low mid high
2 4 5 7 8 9 12 14 17 19 22 25 27 28 33 37
low=mid=high

Binary Search Implementation

Binary Search Analysis

- Each recursive call divides the array in half

If the array is of size n, it divides (and searches) at most $logn$ times before the current half is of size 1

$O(logn)$

Comparable

Binary search on a list of objects requires that the objects have natural ordering

In other words, the objects must implement `Comparable`

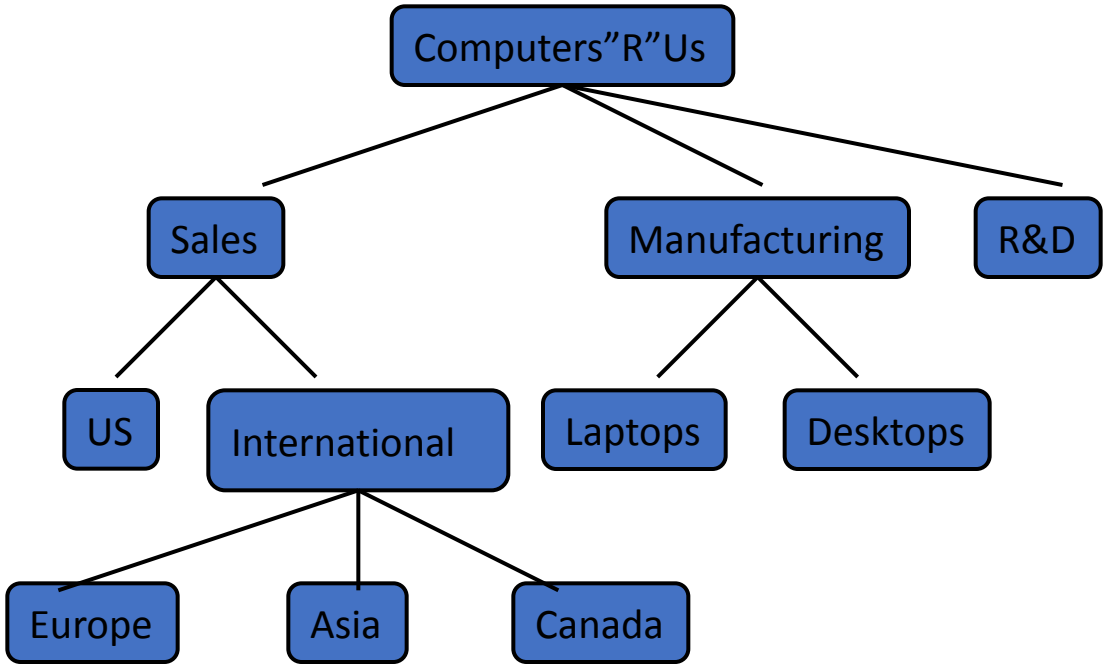
`compareTo`

Tree

A tree is a **hierarchical** data structure

Node: individual elements in the tree

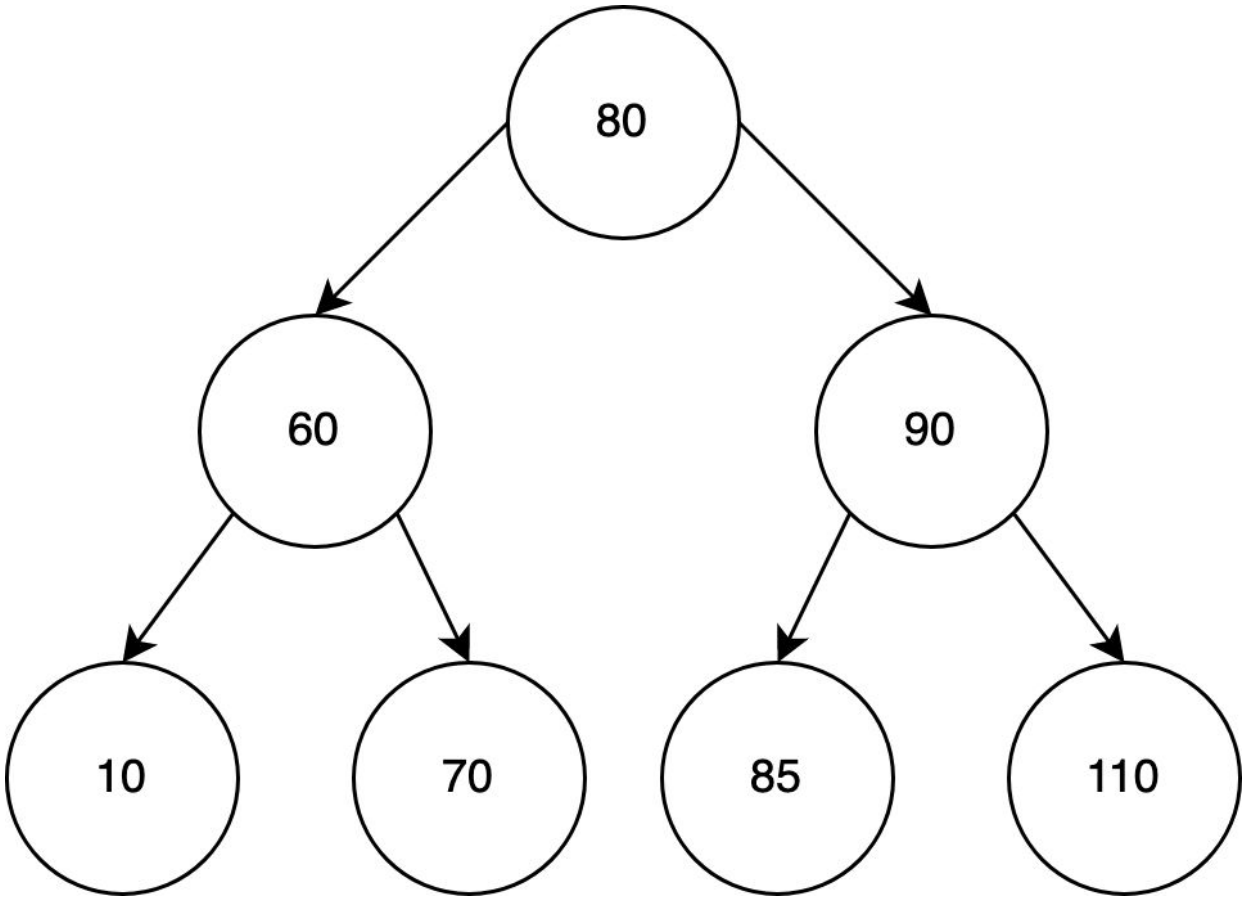
Nodes have a parent-child relation



Trees: Nodes

```
class Node {
    int key;
    Node left;
    Node right;

    public Node(int item) {
        key = item;
        left = null;
        right = null;
    }
}
```



Terminology

root: no parent

A

external/leaf node: no children

E, I, J, K, G, H, D

internal node: - node with at least one child

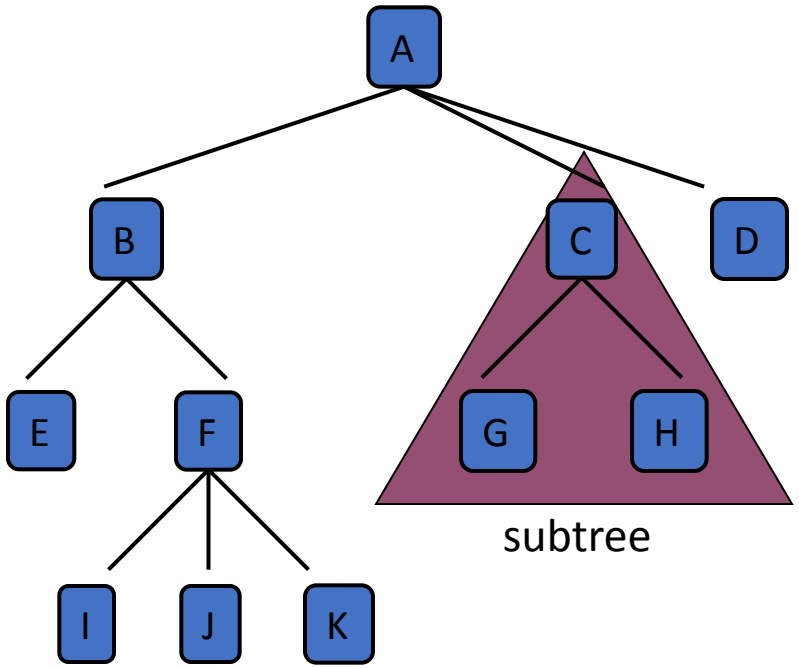
A, B, C, F

parent/child

depth - # of ancestors

Height - Maximum number of edges from a leaf node to the root

- **Subtree**: tree consisting of a node and its descendants



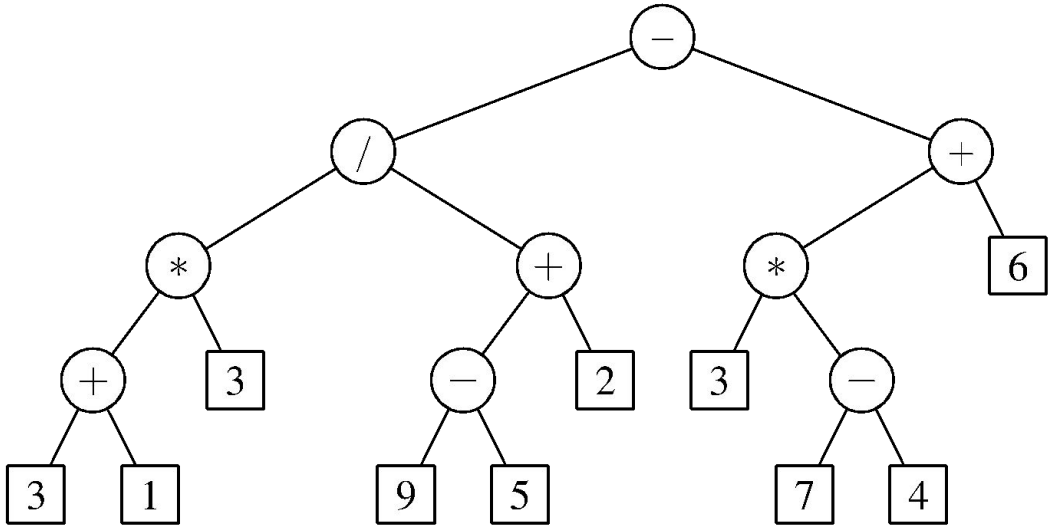
Binary Tree

Each node in a **binary tree** has at most two children

Recursive definition:

Each node has at most two children

- Both subtrees are **binary trees**

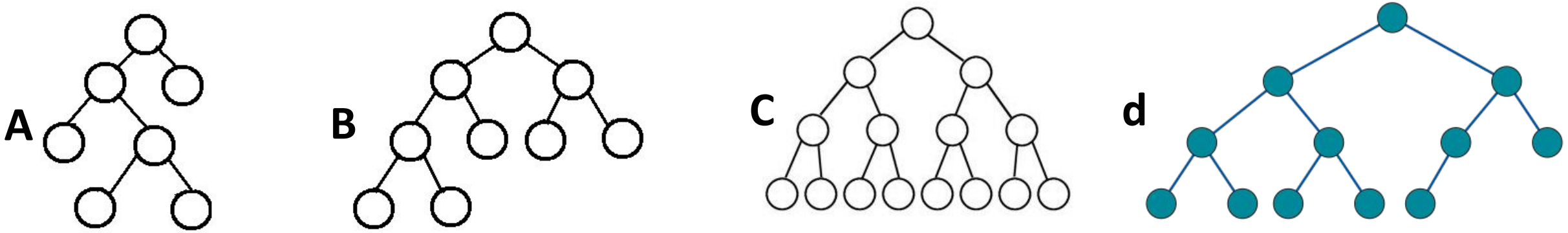


Types of Binary Trees

A binary tree is **full** (or proper) if each node has **zero or two children**

A binary tree is **complete** if every level (except possibly the last) is fully filled and, if the last level of the tree is not fully filled, the nodes of that level are filled from left to right

If a complete binary tree is filled at every level, it is **perfect**



Types of Binary Trees

A binary tree is **full** (or proper) if each node has **zero or two children**

A binary tree is **complete** if every level (except possibly the last) is fully filled and leftmost aligned

If a complete binary tree is filled at every level, it is **perfect**

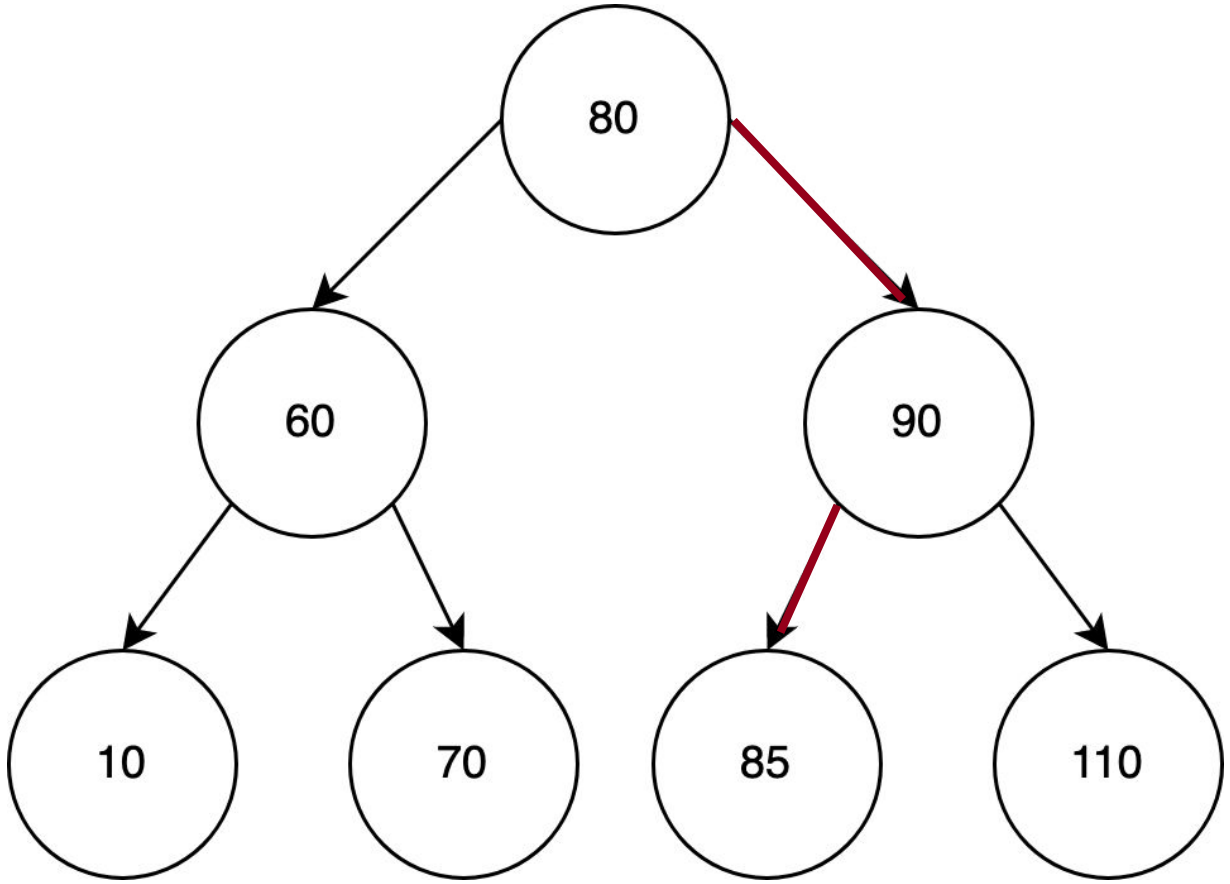
Q1: Is every full binary tree a complete binary tree?

Q2: Is every complete binary tree a full binary tree?
Q3: Is every perfect binary tree a full binary tree?

Binary Trees: Height

Height of a tree:

Maximum number of edges from a leaf node to the root



Height? 2

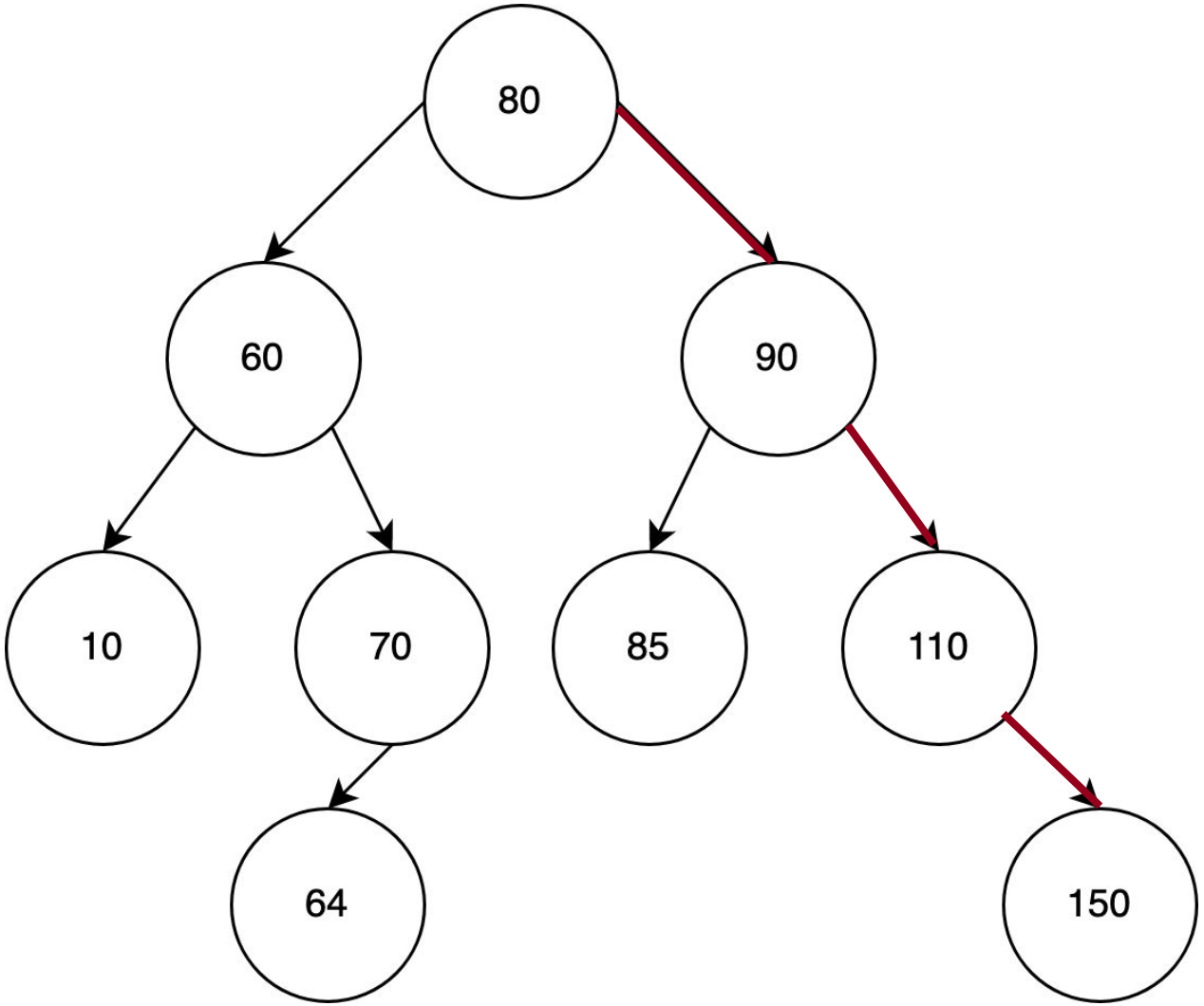
$\log_2(7) \approx 2$

Tree Review

Height? 3

$\log_2(9) \approx 3$

Height of a binary tree is roughly log(n) where n is number of nodes

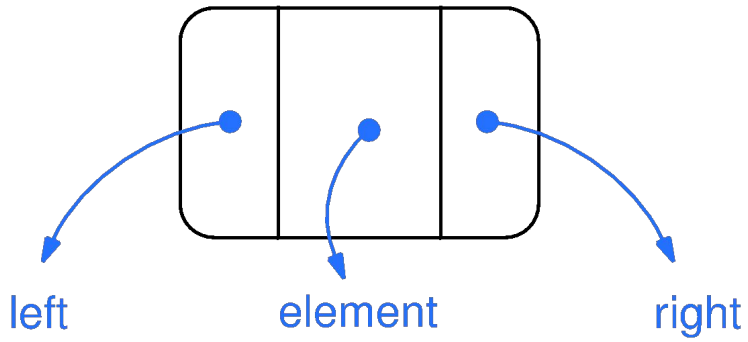


Binary Tree Interface

```
public interface BinaryTree<E extends
Comparable<E>> {
  int size();
  boolean isEmpty();
  void insert(E element);
  boolean contains(E element);
  ...
}
```

Node Implementation

```
public class Node<E> {
  private E element;
  private Node<E> left;
  private Node<E> right;
  //constructors, getters, setters
  }
}
```



Class

```
public class LinkedBinaryTree<E extends
Comparable<E>> implements BinaryTree<E> {
  // what instance variables?
  // nested Node class


}
```

Summary

- iterators
  - What is the **Iterable** interface? iterator()
  - What is the **Iterator** interface?
    - next(), hasNext()

- Binary Search
  - runtime complexity? O(logn)
  - more, less, or equal efficiency to a linear search?
    - O(n)
    - O(logn)